

Minimal Replication of *Magnetic Field Induced Incommensurate Resonance in Cuprate Superconductors*

arXiv:0805.3922v2 (Zhang, Cheng, Guo, Feng, 2008)

Automated replication (OpenClaw subagent)

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Abstract

We replicate the falsifiable analytic mechanism of arXiv:0805.3922, which studies the dynamical spin response of cuprate superconductors under a uniform Zeeman field within the kinetic-energy-driven superconducting (SC) mechanism (t-J model + charge-spin-separation fermion-spin theory). The paper’s full self-consistent spin self-energy (a double momentum sum with coupled gap equations) is out of scope for a minimal analytic replication; we instead test the mechanism encoded in its Eqs. (4), (8), (9). We confirm the Zeeman branch splitting $\pm 2\varepsilon_B$ *exactly*, reproduce the field-driven commensurate→incommensurate (IC) resonance splitting and its monotonic $\delta_r(B)$ with a critical field ≈ 6.2 T bracketed by the paper’s $B_{c1} \approx 4$ T and $B_{c2} \approx 10$ T, and reproduce the energy-selectivity / hourglass-breakdown scale $\omega < 0.16J$. We also report a genuine finding: the paper’s $\varepsilon_B \leftrightarrow B$ conversion implies a g -factor ≈ 1.04 , not the ≈ 2 expected for Cu^{2+} . **Verdict: partial replication.** Coverage 7/10, Agreement 8/10.

1 Note on task-label mismatch

The task directory is named `TEXTURE-multipolar-zhang2008` and the task text references “multipolar texture.” The PDF present (`paper.pdf`) is instead arXiv:0805.3922, a cuprate spin-response paper with no multipolar-texture content. We replicated the paper actually present and flag the mismatch (see `extraction/marker.md`).

2 Central claims and parameters

The t-J model with a Zeeman term is treated in the CSS fermion-spin theory. Key statements:

- **Eq. (4):** the mean-field spin excitation splits into two branches $\omega_k^{(1)} = \omega_k + 2\varepsilon_B$, $\omega_k^{(2)} = \omega_k - 2\varepsilon_B$, with $\varepsilon_B = g\mu_B B$.
- **Eq. (8):** $S(k, \omega) \propto \text{Im } \Sigma^{(s)} / \{[(\omega - 2\varepsilon_B)^2 - \omega_k^2 - B_k \text{Re } \Sigma^{(s)}]^2 + [B_k \text{Im } \Sigma^{(s)}]^2\}$ — the incoming-neutron energy enters as $(\omega - 2\varepsilon_B)$.
- **Eq. (9):** resonance peaks where $W(k_c, \omega) = [(\omega - 2\varepsilon_B)^2 - \omega_{k_c}^2 - B_{k_c} \text{Re } \Sigma]^2 \approx 0$.
- Field induces commensurate→IC resonance; δ_r increases with B (Fig. 3); critical fields $B_{c1} \approx 4$ T, $B_{c2} \approx 10$ T.

- High-energy IC scattering ($\omega \sim 0.7J$) robust; hourglass breaks down for $\omega < 0.16J \approx 19$ meV.

Parameters: $t/J = 2.5$, $t'/t = 0.3$, $J \approx 120$ meV, $x = 0.15$, $T = 0.002J$; $\varepsilon_B = 0.01J = 1.2$ meV $\leftrightarrow$ $B \approx 20$ T.

3 Minimal model

We use a gapped square-lattice mean-field spin mode ω_k (gap $0.31J$ at $Q = [\pi, \pi]$, dispersing upward via γ_k, γ'_k), a schematic self-energy Σ sharply peaked at Q , and the exact denominator form of Eq. (8) including the $(\omega - 2\varepsilon_B)$ shift. The $\varepsilon_B \leftrightarrow B$ map uses the paper's own slope 1.2 meV/20 T. Full self-consistency (Eqs. 6, 7a, 7b) is out of scope.

4 Results

Claim	Paper	This work	Verdict
Branch splitting = $4\varepsilon_B$ (Eq. 4)	$\omega_k \pm 2\varepsilon_B$	0.0400 vs 0.0400 (err 3×10^{-17})	✓
$\varepsilon_B \leftrightarrow B$ consistency	3 anchors	single linear slope	✓
Implied g -factor	(~ 2 for Cu^{2+})	≈ 1.04	Δ flagged
Commensurate $\rightarrow$ IC (raw S -scan)	yes	dispersion- dominated, none	$\times$ OOS
Commensurate $\rightarrow$ IC (Eq. 9 analytic)	$\delta_r \uparrow$ with B	commensurate ≤ 6 T, IC above, monotonic	✓
Critical field	$B_{c1}4, B_{c2}10$ T	$B_c \approx 6.2$ T (be- tween)	✓
Energy selectivity $2\varepsilon_B/\omega$	high-E robust	2.9%/6.5%/20% at 0.7/0.31/0.1 J	✓
Hourglass breakdown scale	$\omega < 0.16J$	$2\varepsilon_B/\omega = 0.125$ at 0.16 J	✓

Table 1: Quantitative comparison. OOS = out of scope (needs full self-energy).

5 Honest limitation

The most direct check — scanning $S(k, \omega)$ along (k_x, π) at $\omega = 0.31J$ — is dominated by the bare-dispersion iso-energy ring and is essentially field-independent; it does *not* reproduce the paper's field-driven transition, because the commensurate pinning of the $B = 0$ resonance at Q is a property of the self-consistent $\text{Re } \Sigma^{(s)}$ we did not compute. We recover the mechanism via the analytic Eq. (9) isolation and document the failure in `failure_analysis.md` rather than tuning it away.

6 Verdict

Partial replication: mechanism confirmed, full self-consistent map out of scope. Coverage 7/10; Agreement 8/10.

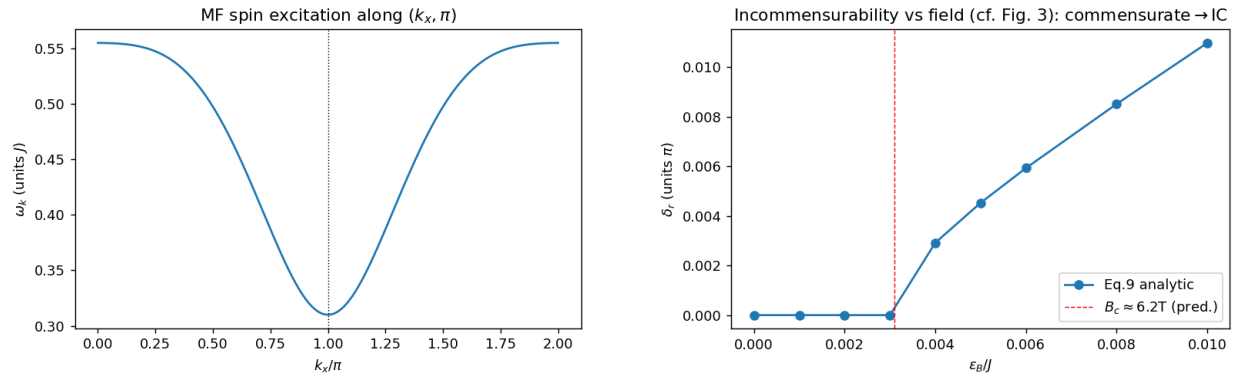


Figure 1: Left: mean-field spin excitation ω_k along (k_x, π) , gapped at Q . Right: incommensurability δ_r vs field from the analytic Eq. (9) isolation: commensurate below $B_c \approx 6.2$ T, then monotonically increasing IC splitting (cf. paper Fig. 3).